Owen Yao

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EDUCATION

Purdue University — *West Lafayette, IN*
Bachelor of Science, Computer Science

EXPERIENCE

Purdue STARS (Summer Training on Awareness and Readiness for Semiconductors) — *West Lafayette, IN*
IC Design Intern — *June - Aug 2025*

- Designed and implemented a 32-bit RISC-V CPU with I Cache and Graphing Peripherals (UART Keypad + SPI LCD Display). Final size is equivalent to 5530/7680 FPGA cells and ~13K standard on taped out NEBULA III chip.
- Wrote full RISC-V assembly (~1200 lines in RAM) implementing graphing calculator functionality, complete with floating point extensions and custom multiply commands.

Atom Bioworks Inc — *RTP, NC*
SWE Intern — *June – Aug 2024*

- Contributed to AptaBLE, an NSF SBIR-funded research project, by building ML pipeline tools for developing a 150M parameters protein language model-based binder classifier and generator, whose accuracy outperformed AlphaFold 3.
- Presented at NeurIPS 2024 conference, with 4497 total accepted papers of 17941 submissions (25.8% acceptance rate). Poster sections: AI for New Drug Modalities, Machine Learning in Structural Biology.

Lenovo Inc — *Morrisville, NC*
SWE Intern — *June - Aug 2024*

- Built a Python tool to automate MI data analysis and chart generation for use by Geo C-level leadership team in weekly reviews, reducing turnaround time by 92%.
- Created proof-of-concept Large Language Models (LLMs) demos and evaluated Table Question Answering capability of top table-trained LLMs to guide future plan for enabling query of market data.

Department of Material Science Engineering (MSE), NC State University — *Raleigh, NC*
SWE Intern — *June – Aug 2023*

- Worked with NCSU undergraduates and postgraduates at the NCSU MSE lab to investigate the ability of memristors to solve non-linear problems by analyzing simulations of memristive networks and circuits.
- Researched effect of voids on percolation; generated void representatives on a network simulation [Percolator and CircuitSymphony] and visualized data using 3D graphical analysis.

PUBLICATIONS

- Patel, Sawan. Peng, Zhangzhi, Friedman, Adam, Yao, Owen. Fraser, Keith. Chatterjee, Pranam. AptaBLE As A Deep Learning Platform for SELEX Optimization. Presented poster at “AI for New Drug Modalities” workshop at NeurIPS 2024.
- Frick, Nikolai. LaBean, Thomas. Trost, Isaac. Yao, Owen. Exploration of Three Dimensional Neuromemristive Material. Presented at: NC State Undergraduate Research and Creativity Summer Symposium. July 27th, 2023. D. H. Hill Jr. Library.

SKILLS

- **Software languages:** Python, Java, C/C++, R
- **Software tools:** PyTorch, Scikit-learn, Pandas, Seaborn, Linux, Unix, Git, HuggingFace
- **IC design:** SystemVerilog, GTKWave, logic analyzer/oscilloscope, breadboarding